

SOTA Models (0.5B–7B) for English/Russian NLP Tasks

Model (HF ID)	Params	Languages/ Tasks	Strengths & Use- Cases	Weaknesses
LLaMA-2-7B-Chat (meta-llama/Llama-2-7b-chat)	7B	English (dialogue, generation)	Strong general-purpose LLM optimized for chat. Outperforms most open-source chat models on benchmarks and matches closed models like GPT and PaLM in helpfulness/safety ¹ . Good few-shot performance.	Still prone to hallucinations and knowledge cut-off (trained by mid-2023). Limited to training data; not expert-level.
Mistral-7B (mistralai/Mistral-7B)	7.3B	English (text/ code generation)	Highest-quality 7B model. Beats LLaMA-2-13B on almost all tasks ² ³ , with especially strong reasoning and code generation. Uses GQA/SWA for speed. Apache-2.0 license (commercial use).	Still English-centric; smaller parameter count limits world knowledge (on par with larger models only on knowledge tasks) ³ .
Falcon-7B-Instruct (tiiuae/falcon-7b-instruct)	7B	English (instruct/chat)	High performance among 7B models. Trained on 1.5T tokens (RefinedWeb+curated data). Outperforms comparable 7B open models (MPT-7B, StableLM, RedPajama) ⁴ . Optimized for fast inference (FlashAttention, multiqueue). Apache-2.0 license.	Mostly trained on English (and some French); <i>does not generalize well to other languages</i> ⁵ . Exhibits typical LLM biases from web data.

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MPT-7B (Base) (mosaicml/mpt-7b)	6.7B	English (text generation, long-context)	Matches or exceeds LLaMA-7B quality ⁶ . Trained on 1T tokens (text+code), Apache-2.0. Specially designed for very long contexts (up to ~65K tokens via ALiBi) ⁷ . Efficient training/inference (FlashAttention).	English-centric; no instruction fine-tuning by default (base model). Early 2023 model, so may lack newest knowledge.
OPT-6.7B (facebook/opt-6.7b)	6.7B	English (text generation, few-shot)	Openly released “GPT-3 class” model ⁸ . Good at zero/few-shot generation and can be fine-tuned. Relatively fast with half-precision.	Pretrained only on data up to 2020 (no RLHF). Lacks instruction tuning (though INSTRUCT variants exist). License is Meta’s research license (“other”).
GPT-J-6B (EleutherAI/gpt-j-6B)	6B	English (text generation)	Among earliest open 6B LMs; performs well on generation tasks given prompts. Open license (Apache-2.0).	Older model (2021); weaker than newer 7B models (LLaMA-2, Mistral). No fine-tuned instruction variant in HF (community forks exist).

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BLOOM-7B1 (bigscience/bloom-7b1)	7.1B	Multilingual (46 languages)	Huge language coverage (45+ languages) ⁹ . Good for multilingual generation/translation, including Russian. RAIL license (non-commercial use only).	Lower performance per parameter than specialized models (it was trained with a smaller compute budget). License restricts commercial use.
ruGPT3-Large (ai-forever/rugpt3large_based_on_gpt2)	760M	Russian (text generation)	Open Russian GPT-like model (760M) ¹⁰ . Good fluent Russian generation and dialogue. Outperforms smaller RuBERT for generation tasks. MIT license (commercial use).	Moderate size – knowledge limited. Behaves like GPT-2 architecture (no instruction tuning). May hallucinate or produce unsafe Russian content.
ruT5-Large (ai-forever/ruT5-large)	737M	Russian (summarization, QA, translation)	Encoder-decoder model for Russian, similar to T5. Good at generation tasks (summaries, Q/A, translation). Pretrained on Russian corpora ¹¹ . MIT license.	Lacks very long-context ability. Outperformed by larger models (like FRED-T5-XL) on some tasks ¹² .

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FRED-T5-Large (ai-forever/FRED-T5-large)	820M	Russian (summarization, generation)	Very strong Russian seq2seq model. Trained on diverse “Mixture of Denoisers” objective ¹¹ . Often achieves higher accuracy than ruT5-large on reading comprehension and generation tasks ¹³ . MIT license.	Larger (820M) – requires more compute. Still not as versatile as 7B-level LLMs (but those do not exist monolingual in Russian yet).
ruRoBERTa-Large (DeepPavlov)	~550M	Russian (classification, NER)	Large masked LM for Russian (24 layers, ~355M ¹⁴). Good for classification, NER, and feature extraction. Multilingual XLM-R-large (550M) can also be used across RU/EN.	No obvious license issues (Apache/MIT typically). Models from ai-forever or DeepPavlov may lack state-of-art fine-tuning.
XLM-RoBERTa-Large (xlm-roberta-large)	550M	Multilingual (100+ langs)	Strong multilingual encoder (100 languages including RU, EN). State-of-art for classification/NER on many languages. Supports cross-lingual tasks.	Only 512 token limit. Encoder-only (no generation). Large GPU memory for training/inference.
NLLB-200-3.3B (facebook/nllb-200-3.3B)	3.3B	Multilingual (200 langs) MT	State-of-the-art translation model. Achieves excellent accuracy on low-resource directions (e.g. English↔Russian) ¹⁵ . Open-sourced by Meta with permissive license. Ideal for high-quality machine translation.	Large and slow to run. Inference requires cross-attention (slower than decoder-only). Still weaker on extremely long or very informal text.

Model (HF ID)	Params	Languages/ Tasks	Strengths & Use- Cases	Weaknesses
Giga-Embeddings-Instruct (ai-sage/Giga-Embeddings-instruct)	3B	English/Russian (semantic search)	Instruction-tuned dual-encoder for retrieval. Based on GigaChat-3B, with latent-attention pooling ¹⁶ . Produces 2048-dim embeddings. Excels at retrieval/QA tasks (MTEB benchmarks).	Large (3B); slower than light embedding models. Requires careful prompting (instruction). Relatively new (emerging performance claims).
Giga-Retrieval-Instruct (ai-sage/Giga-Retrieval-instruct)	3B	English/Russian (question answering)	Similar to Giga-Embeddings but fine-tuned for QA retrieval ¹⁷ . Generates instructions-aware embeddings (latent-attention) for queries and passages ¹⁸ . Good for RAG pipelines, semantic search.	Same as Giga-Embeddings: large 3B footprint. Must prepend task-specific prompts to get best results ¹⁹ .

Sources: Model details and capabilities are drawn from official model cards and publications ¹ ² ²⁰ ²¹ ¹⁶ ¹⁸ ⁶.

Kaggle “Non-Standard” Text-Generation Solutions

In Kaggle text-generation competitions, top teams often employ creative techniques beyond off-the-shelf LMs. Below are example strategies and published notes on their use in real contests.

Method	Description / Kaggle Examples
Pseudo-Labeling (Self-Training)	Use the model to label unlabeled (test) data and retrain. Kagglers iteratively add high-confidence model outputs as new training data ²² . E.g. multi-round PL with thousands of pseudo-samples improved leaderboards ²³ ²⁴ .
Meta Pseudo-Labeling (MPL)	An enhanced PL where the “teacher” model’s pseudo-labels are optimized based on student validation performance ²⁴ . Kaggle writeups note MPL can yield better pseudo-labels and boost CV/LB scores (inspired by [Pham et al. 2021] ²⁴).

Method	Description / Kaggle Examples
Adversarial Weight Perturbation (AWP)	During fine-tuning, add small perturbations to model weights to improve robustness. Kagglers cite Wu et al. (2020) showing AWP “improves generalization” ²⁵ . Teams report CV gains (e.g., +0.001–0.01) by using adversarial training ²⁶ ²⁵ .
Masked Language Model (MLM) Pretraining	Continue pretraining (or jointly train) with masked token prediction on domain data. Common in Kaggle solutions: e.g. mask 10–20% tokens of task-specific text (sometimes with CutMix augmentation) to improve representation ²⁷ . This was used in competitions like Feedback Prize (MLM + sequence shuffling) ²⁷ .
Model Ensembling	Combine multiple models’ outputs. Kagglers use <i>large ensembles</i> (dozens of models) via weighted averaging or advanced fusion. For example, one team stacked 19 different models and optimized weights (even allowing negative weights) via Nelder–Mead simplex ²⁸ . Others used Weighted Box Fusion (WBF) to merge token probability outputs across models ²⁹ . These ensembles significantly boost final scores.
CutMix / Mixup for Text	Adapted from vision, Kagglers “cut” or swap segments between two text examples to create new training samples. One solution combined MLM masking (15%) with a “text CutMix” (splice one sequence into another) during training ²⁷ . This augmentation increased data diversity and improved robustness on tasks like writing scoring.
Knowledge Distillation	Distill a large ensemble or teacher into a smaller model. Some solutions trained a “student” model using soft targets from a multi-model ensemble or from augmented data ²³ . For example, one team trained a compact model on 4K human essays + 11K “weakly labeled” essays, using multi-round PL and distillation from an ensemble ²³ .

Notes: These strategies often overlap. For instance, ensembles are paired with distilled single-model submissions, and pseudo-labeling is iterated with adversarial finetuning. References indicate Kaggle analyses discussing these methods ²² ²⁷ ²⁵ ³⁰ ²⁴. The exact impact varies by competition, but they reflect the “mix-and-match” approach winners use when tackling generative text tasks.

¹ meta-llama/Llama-2-7b-chat · Hugging Face

<https://huggingface.co/meta-llama/Llama-2-7b-chat>

² ³ Mistral 7B | Mistral AI

<https://mistral.ai/news/announcing-mistral-7b>

⁴ ⁵ tiuae/falcon-7b-instruct · Hugging Face

<https://huggingface.co/tiuae/falcon-7b-instruct>

⁶ ⁷ Introducing MPT-7B: A New Standard for Open-Source, Commercially Usable LLMs | Databricks Blog

<https://www.databricks.com/blog/mpt-7b>

⁸ facebook/opt-6.7b · Hugging Face

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<https://huggingface.co/bigscience/bloom-7b1>

10 11 12 13 14 21 A Family of Pretrained Transformer Language Models for Russian

<https://arxiv.org/html/2309.10931v2>

15 Models – Hugging Face

<https://huggingface.co/models?language=kbp>

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<https://huggingface.co/ai-sage/Giga-Embeddings-instruct>

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<https://huggingface.co/ai-sage/Giga-Retrieval-instruct>

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